DSP_POWER
File: DSP_POWER.kicad_sch

DSP-I
File: DSP-I.kicad_sch

DSP-II
File: DSP-II.kicad_sch

Serial_SATA
File: Serial_SATA.kicad_sch

SD_Card_LCD
File: SD_Card_LCD.kicad_sch

Video_Decoder (page 12)
File: Video_Decoder.kicad_sch

Power_Supply-I
File: Power_Supply-I.kicad_sch

Power_Supply_II
File: Power_Supply_II.kicad_sch

LCD_II
File: LCD_II.kicad_sch

DDR_RAM
File: DDR_RAM.kicad_sch

NAND_Flash
File: NAND_Flash.kicad_sch

Sound_Coded
File: Sound_Coded.kicad_sch

LAN_8710A
File: LAN_8710A.kicad_sch

Connectors
File: Connectors.kicad_sch

Mislleneous
File: Mislleneous.kicad_sch

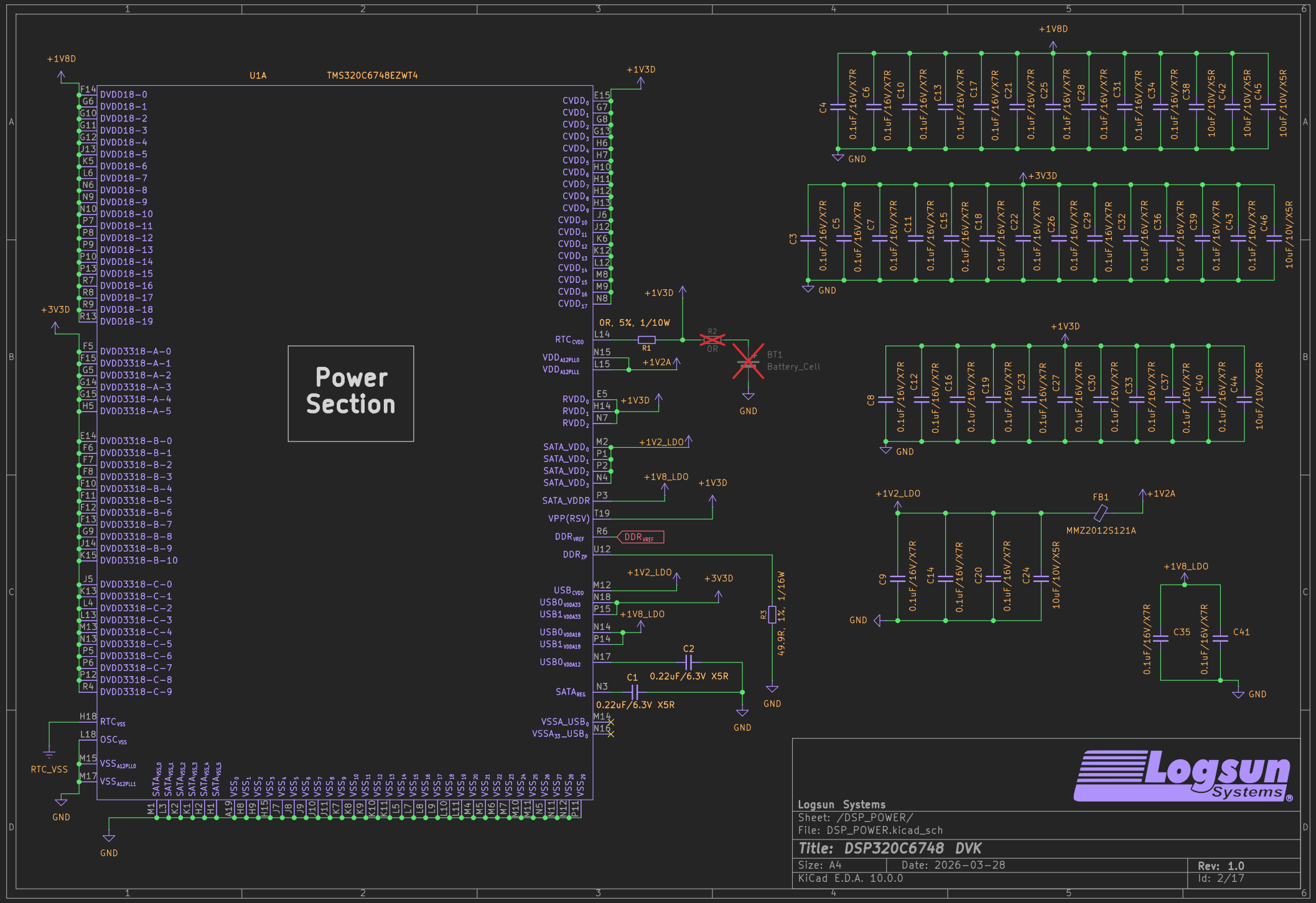
ADC_RTC
File: ADC_RTC.kicad_sch

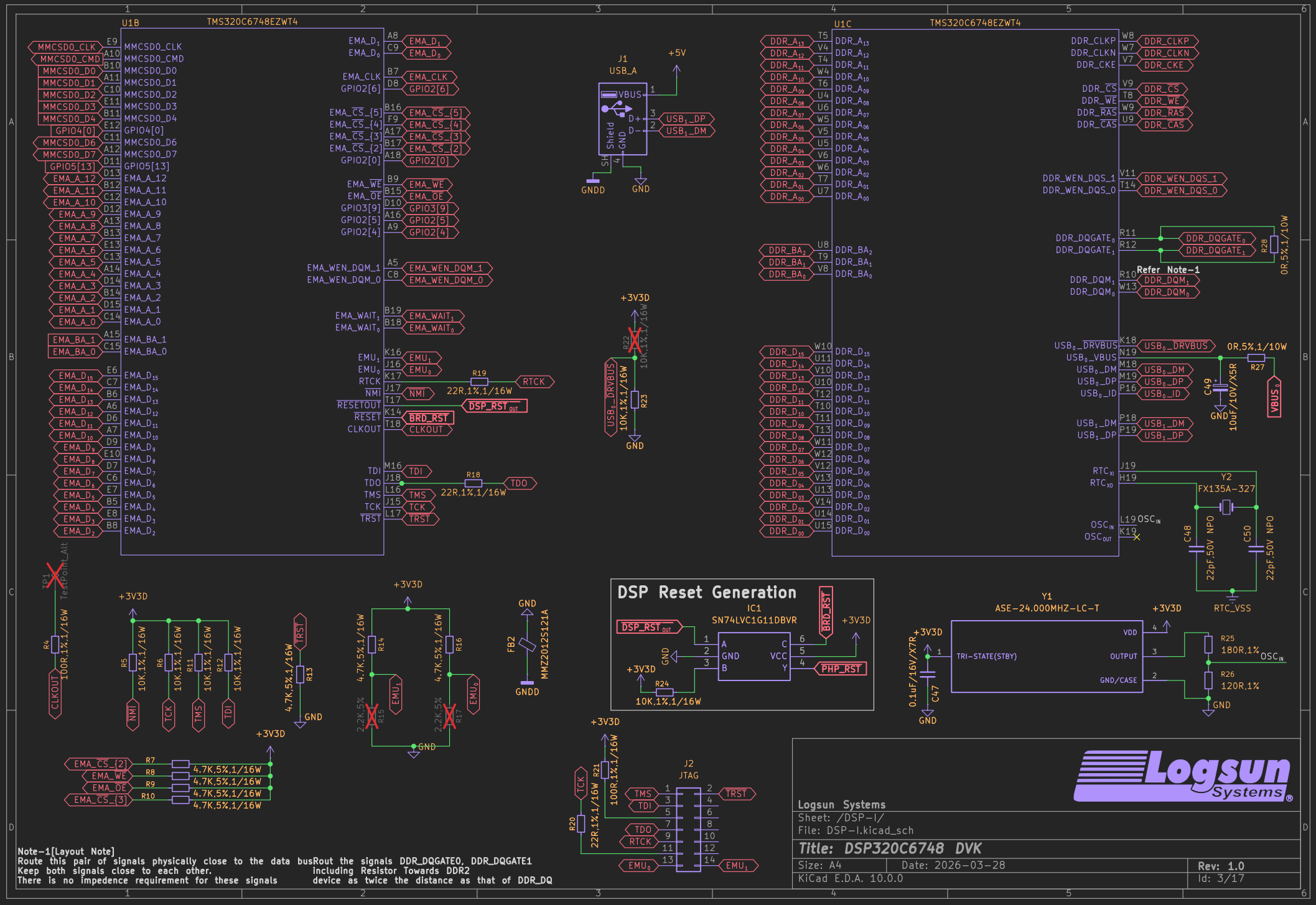
Logsun Systems

Sheet: /
File: DSP320C6748_DVK.kicad_sch

Title: DSP320C6748 DVK

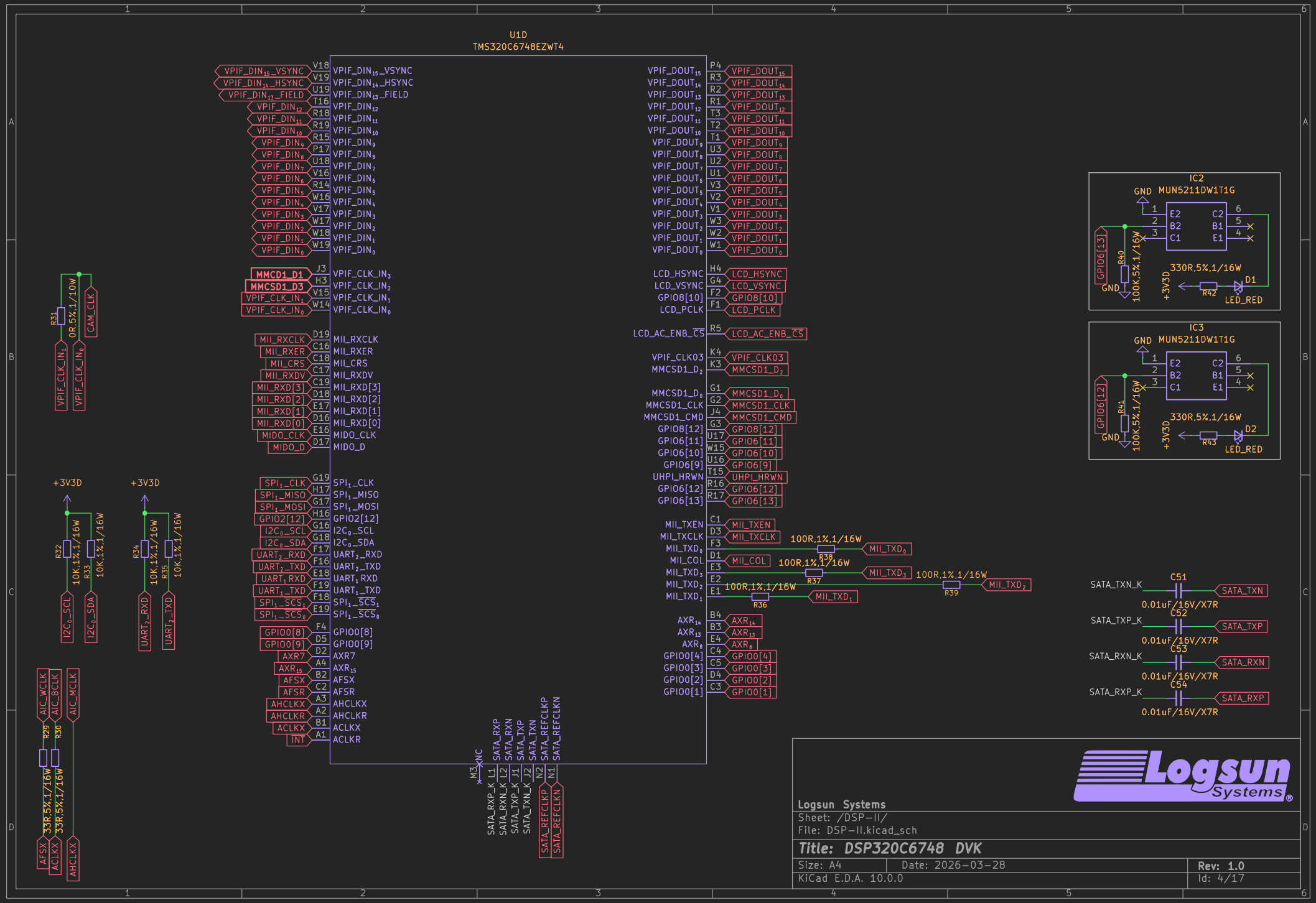
Size: A4	Date: 2026-03-28	Rev: 1.0
KiCad E.D.A. 10.0.0		Id: 1/17

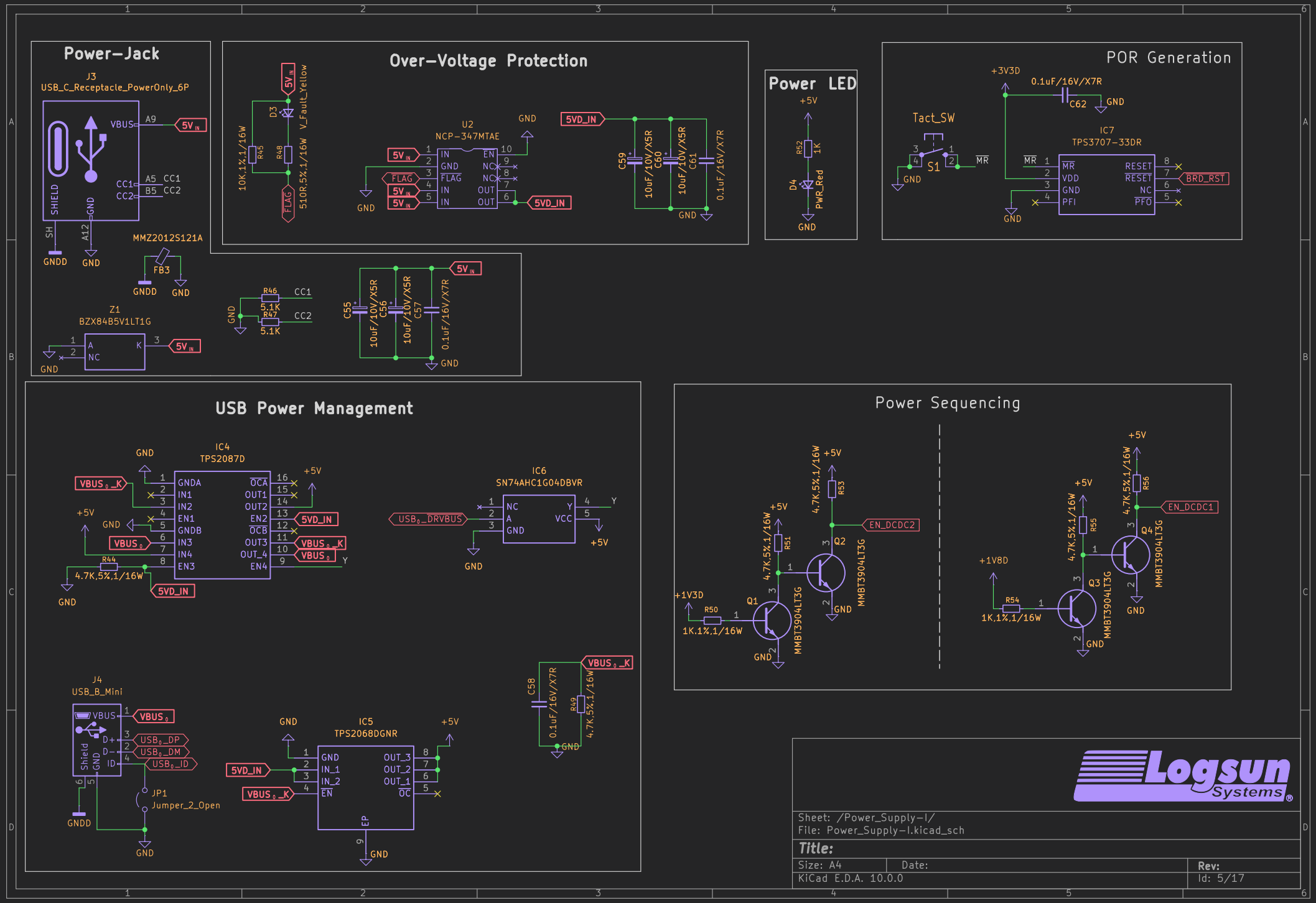

Power Section
U1A
TMS320C6748EZWT4
+1V8D
+3V3D
+1V3D
+1V2A
+1V2_LDO
+1V8_LDO
GND
RTC_VSS
OR, 5%, 1/10W
R1
R2
OR
BT1
Battery_Cell
DDR_VREF
0.22uF/6.3V X5R
C1
C2
R3
49.9R, 1%, 1/16W
FB1
MMZ2012S121A
Logsun Systems
Sheet: /DSP_POWER/
File: DSP_POWER.kicad_sch
Title: DSP320C6748 DVK
Size: A4
Date: 2026-03-28
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 2/17

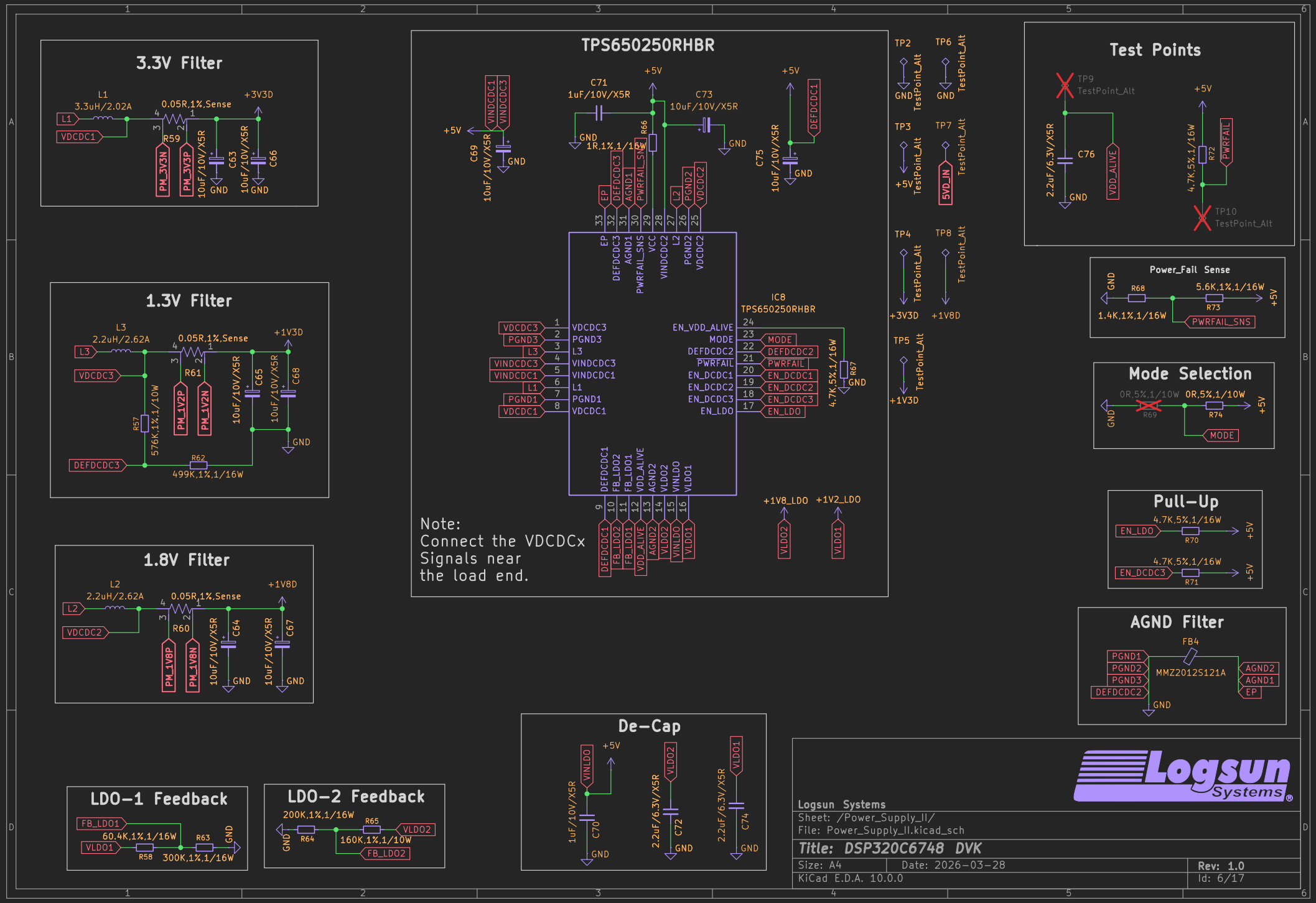


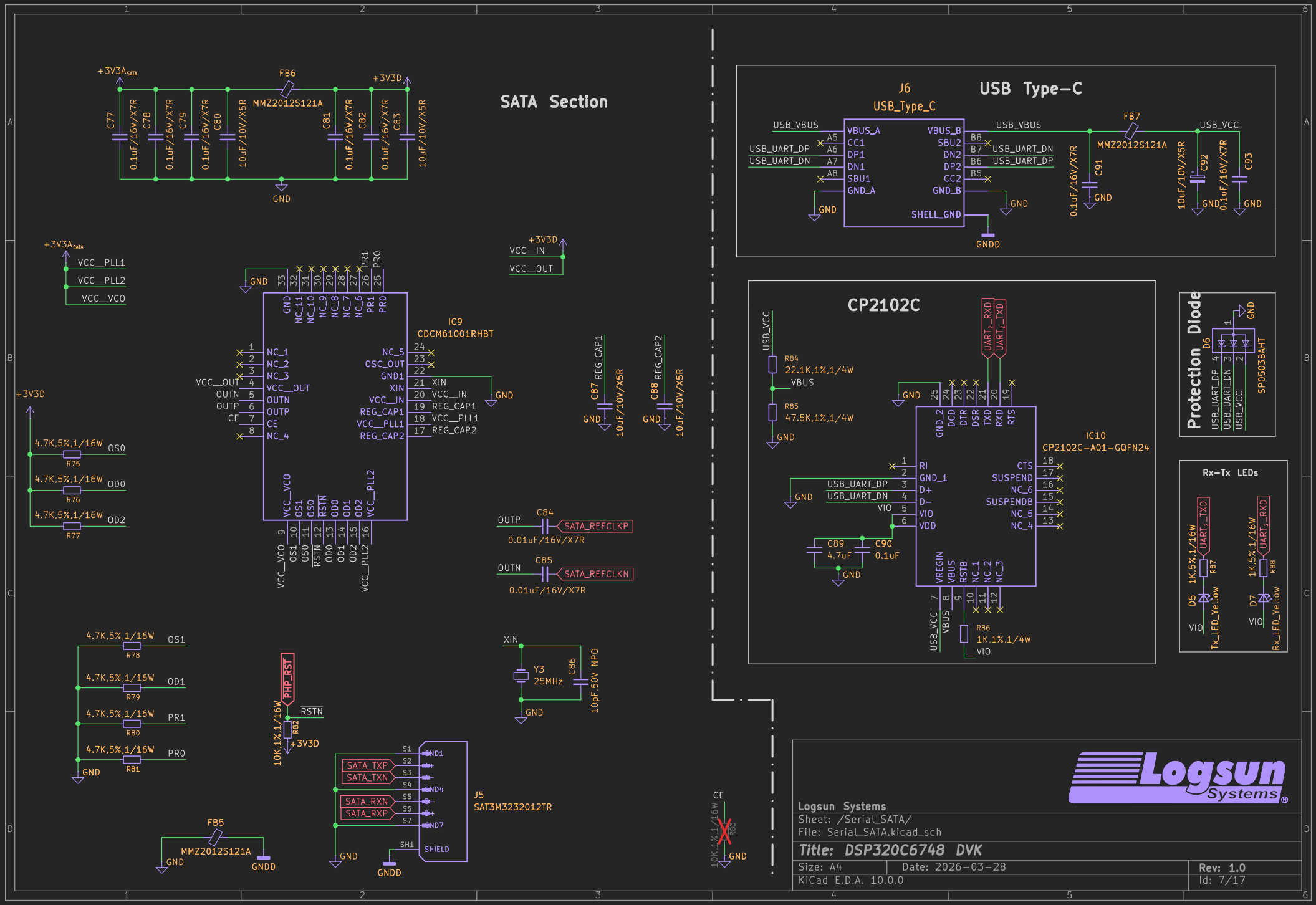
Note-1[Layout Note]
Route this pair of signals physically close to the data bus
Keep both signals close to each other.
There is no impedence requirement for these signals

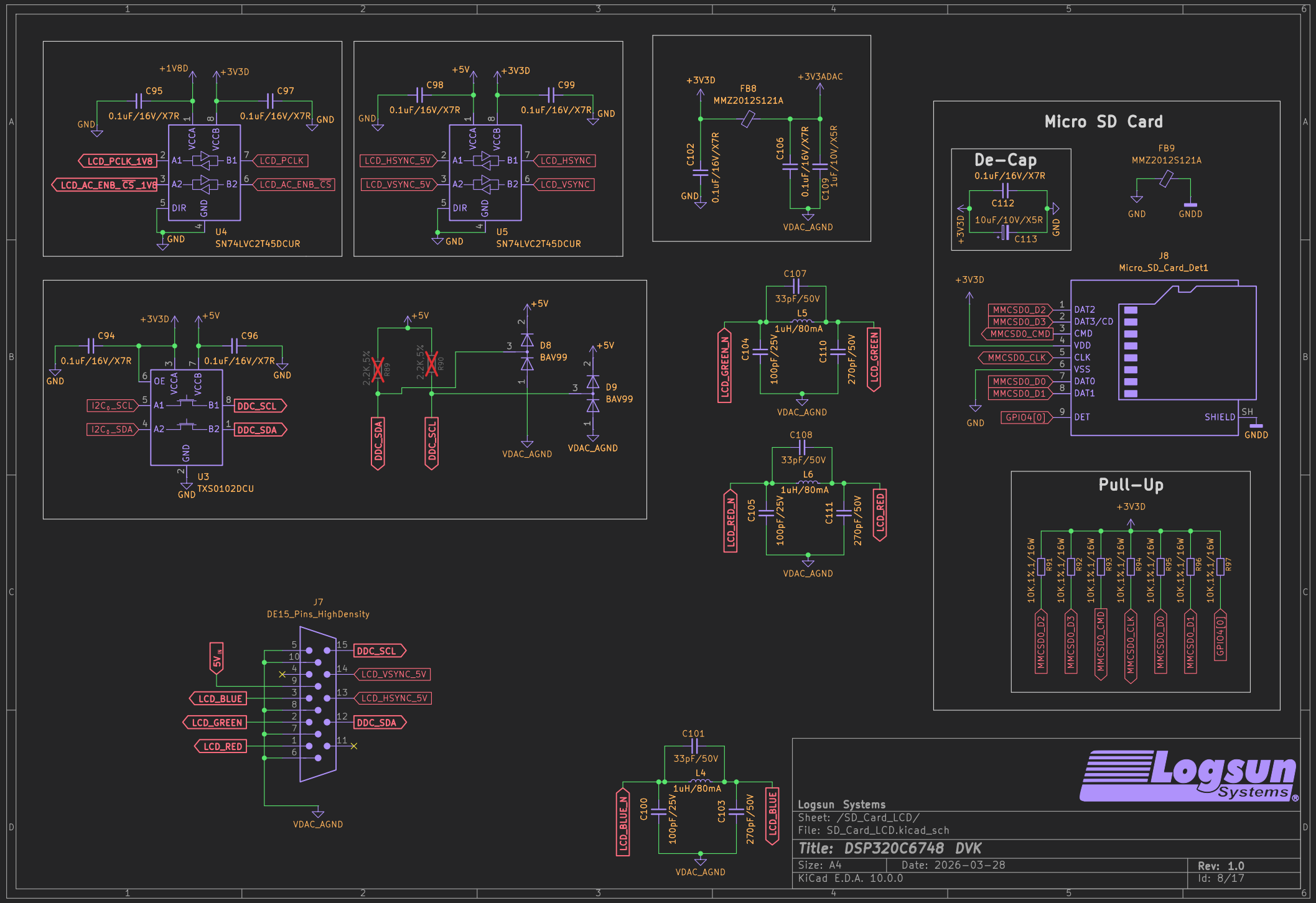
Rout the signals DDR_DQGATE0, DDR_DQGATE1 including Resistor Towards DDR2 device as twice the distance as that of DDR_DQ

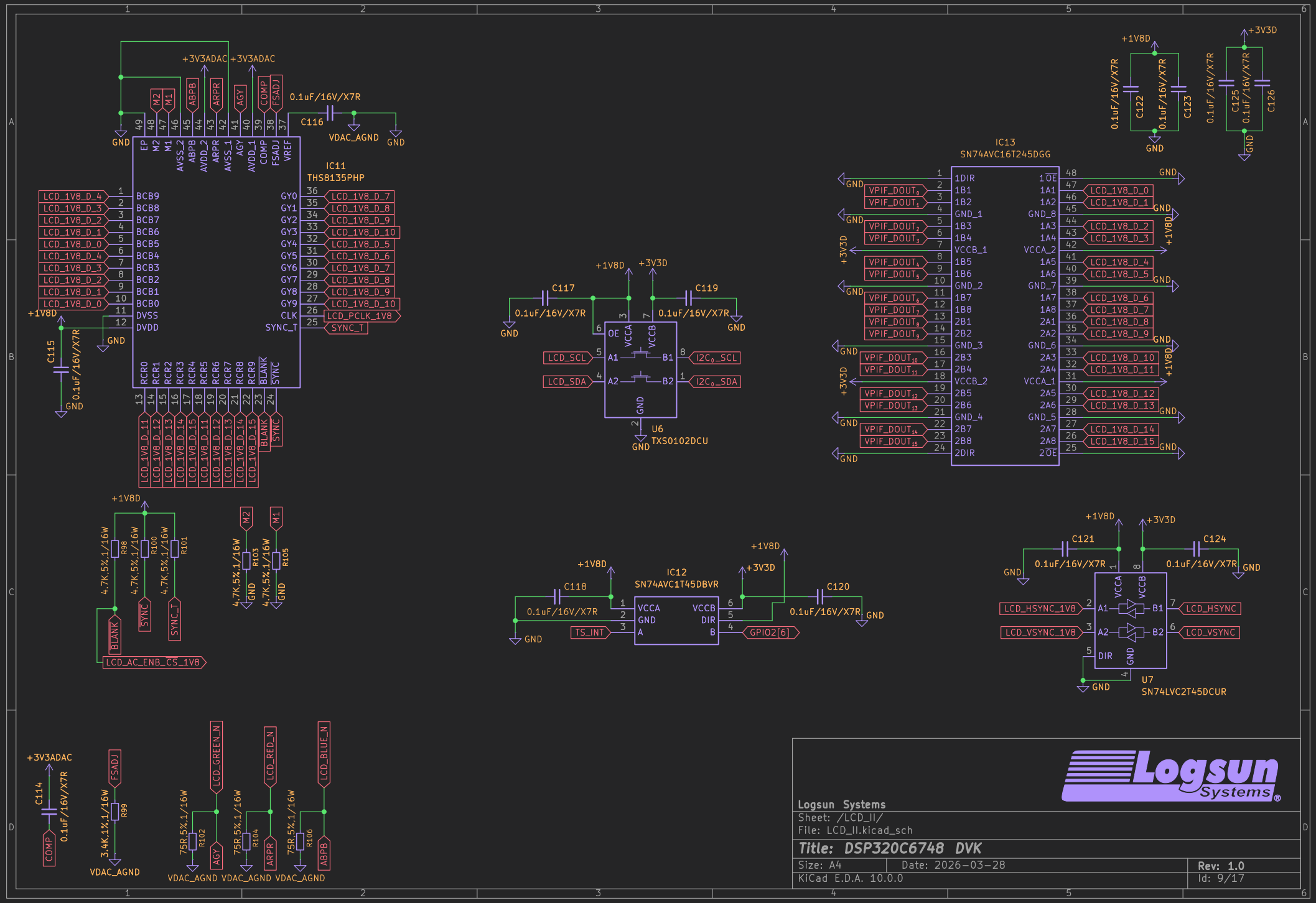

U1D
TMS320C6748EZWT4
IC2
MUN5211DW1T1G
IC3
MUN5211DW1T1G
330R,5%,1/16W
LED_RED
100R,1%,1/16W
10K,1%,1/16W
33R,5%,1/16W
0.01uF/16V/X7R
+3V3D
GND
Logsun Systems
Sheet: /DSP-II/
File: DSP-II.kicad_sch
Title: DSP320C6748 DVK
Size: A4
Date: 2026-03-28
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 4/17

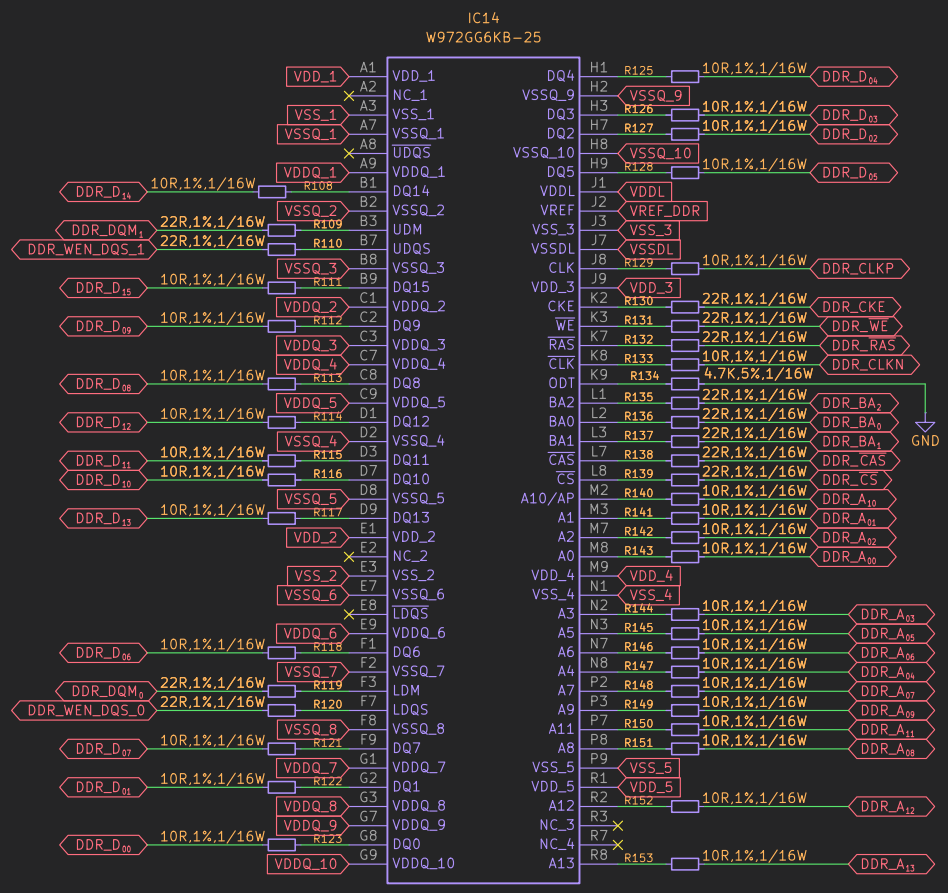

Power-Jack
J3
USB_C_Receptacle_PowerOnly_6P
VBUS
A9
5V_IN
A5 CC1
B5 CC2
CC1
CC2
SHIELD
GND
SH
A12
GNDD
GND
MMZ2012S121A
FB3
GNDD
GND
Z1
BZX84B5V1LT1G
A
NC
K
5V_IN
GND
Over-Voltage Protection
5V_IN
D3
V_Fault_Yellow
10K,1%,1/16W
R45
R48
510R,5%,1/16W
FLAG
U2
NCP-347MTAE
IN
GND
FLAG
IN
EN
NC
NC
OUT
OUT
5VD_IN
GND
C59
10uF/10V/X5R
C60
10uF/10V/X5R
C61
0.1uF/16V/X7R
R46
5.1K
CC1
R47
5.1K
CC2
C55
10uF/10V/X5R
C56
10uF/10V/X5R
C57
0.1uF/16V/X7R
Power LED
+5V
R52
1K
D4
PWR_Red
GND
POR Generation
+3V3D
0.1uF/16V/X7R
C62
GND
Tact_SW
S1
MR
IC7
TPS3707-33DR
MR
VDD
GND
PFI
RESET
RESET
NC
PFO
BRD_RST
USB Power Management
IC4
TPS2087D
GNDA
IN1
IN2
EN1
GNDB
IN3
IN4
EN3
OCA
OUT1
OUT2
EN2
OCB
OUT3
OUT_4
EN4
VBUS0_K
VBUS0
5VD_IN
R44
4.7K,5%,1/16W
IC6
SN74AHC1G04DBVR
USB0_DRVBUS
NC
A
GND
VCC
Y
J4
USB_B_Mini
VBUS
D+
D-
ID
Shield
GND
USB0_DP
USB0_DM
USB0_ID
JP1
Jumper_2_Open
IC5
TPS2068DGNR
GND
IN_1
IN_2
EN
OUT_3
OUT_2
OUT_1
OC
EP
C58
0.1uF/16V/X7R
R49
4.7K,5%,1/16W
Power Sequencing
+5V
4.7K,5%,1/16W
R53
EN_DCDC2
Q2
MMBT3904LT3G
R51
Q1
+1V3D
R50
1K,1%,1/16W
R55
Q3
+1V8D
R54
1K,1%,1/16W
R56
EN_DCDC1
Q4
Logsun Systems
Sheet: /Power_Supply-1/
File: Power_Supply-1.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.0
Id: 5/17

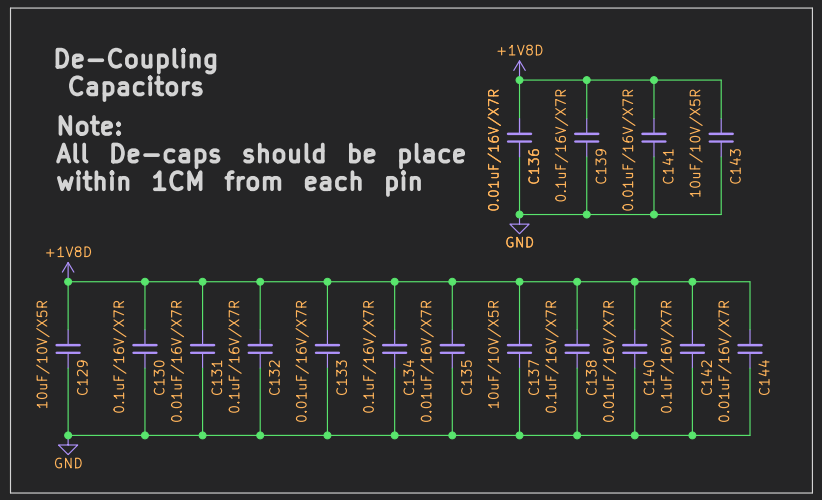

3.3V Filter
L1
3.3uH/2.02A
0.05R,1%,Sense
+3V3D
L1
VDCDC1
R59
PM_3V3N
PM_3V3P
10uF/10V/X5R
C63
C66
GND
1.3V Filter
L3
2.2uH/2.62A
0.05R,1%,Sense
+1V3D
L3
VDCDC3
R61
PM_1V2P
PM_1V2N
C65
C68
R57
576K,1%,1/10W
DEFDCDC3
R62
499K,1%,1/16W
1.8V Filter
L2
2.2uH/2.62A
0.05R,1%,Sense
+1V8D
L2
VDCDC2
R60
PM_1V8P
PM_1V8N
C64
C67
TPS650250RHBR
C71
1uF/10V/X5R
C73
10uF/10V/X5R
C69
10uF/10V/X5R
R66
1R,1%,1/16W
C75
10uF/10V/X5R
IC8
TPS650250RHBR
VDCDC3
PGND3
L3
VINDCDC3
VINDCDC1
L1
PGND1
VDCDC1
EN_VDD_ALIVE
MODE
DEFDCDC2
PWRFAIL
EN_DCDC1
EN_DCDC2
EN_DCDC3
EN_LDO
R67
4.7K,5%,1/16W
EP
DEFDCDC3
AGND1
PWRFAIL_SNS
VCC
VINDCDC2
L2
PGND2
VDCDC2
DEFDCDC1
FB_LDO2
FB_LDO1
VDD_ALIVE
AGND2
VLDO2
VINLDO
VLDO1
+1V8_LDO
+1V2_LDO
Note:
Connect the VDCDCx
Signals near
the load end.
TP2
TP3
TP4
TP5
TP6
TP7
TP8
TestPoint_Alt
+3V3D
+1V8D
+1V3D
5VD_IN
Test Points
TP9
TP10
C76
2.2uF/6.3V/X5R
VDD_ALIVE
R72
4.7K,5%,1/16W
PWRFAIL
Power_Fail Sense
R68
1.4K,1%,1/16W
R73
5.6K,1%,1/16W
PWRFAIL_SNS
Mode Selection
0R,5%,1/10W
R69
R74
MODE
Pull-Up
4.7K,5%,1/16W
EN_LDO
R70
EN_DCDC3
R71
AGND Filter
FB4
MMZ2012S121A
PGND1
PGND2
PGND3
DEFDCDC2
AGND2
AGND1
EP
De-Cap
VINLDO
VLDO2
VLDO1
C70
C72
C74
LDO-1 Feedback
FB_LDO1
VLDO1
60.4K,1%,1/16W
R58
300K,1%,1/16W
R63
LDO-2 Feedback
200K,1%,1/16W
R64
R65
160K,1%,1/10W
VLDO2
FB_LDO2
Logsun Systems
Sheet: /Power_Supply_II/
File: Power_Supply_II.kicad_sch
Title: DSP320C6748 DVK
Size: A4
Date: 2026-03-28
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 6/17

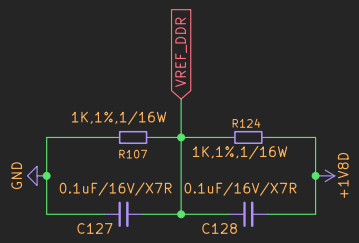

SATA Section
USB Type-C
CP2102C
Protection Diode
Rx-Tx LEDs
IC9
CDCM61001RHBT
J6
USB_Type_C
IC10
CP2102C-A01-GQFN24
J5
SAT3M3232012TR
FB6
MMZ2012S121A
FB7
FB5
Y3
25MHz
+3V3A_SATA
+3V3D
GND
GNDD
Logsun Systems
Sheet: /Serial_SATA/
File: Serial_SATA.kicad_sch
Title: DSP320C6748 DVK
Size: A4
Date: 2026-03-28
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 7/17

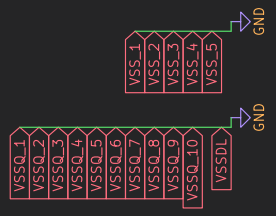

+1V8D
+3V3D
C95
0.1uF/16V/X7R
C97
0.1uF/16V/X7R
GND
LCD_PCLK_1V8
LCD_AC_ENB_CS_1V8
LCD_PCLK
LCD_AC_ENB_CS
VCCA
VCCB
A1
A2
B1
B2
DIR
U4
SN74LVC2T45DCUR
+5V
C98
0.1uF/16V/X7R
C99
0.1uF/16V/X7R
LCD_HSYNC_5V
LCD_VSYNC_5V
LCD_HSYNC
LCD_VSYNC
U5
SN74LVC2T45DCUR
+3V3D
FB8
MMZ2012S121A
+3V3ADAC
C102
0.1uF/16V/X7R
C106
0.1uF/16V/X7R
C109
1uF/10V/X5R
VDAC_AGND
C94
0.1uF/16V/X7R
C96
0.1uF/16V/X7R
OE
I2C0_SCL
I2C0_SDA
DDC_SCL
DDC_SDA
U3
TXS0102DCU
2.2K 5%
R89
R90
D8
BAV99
D9
BAV99
C107
33pF/50V
L5
1uH/80mA
C104
100pF/25V
C110
270pF/50V
LCD_GREEN_N
LCD_GREEN
C108
33pF/50V
L6
1uH/80mA
C105
100pF/25V
C111
270pF/50V
LCD_RED_N
LCD_RED
J7
DE15_Pins_HighDensity
5V_IN
LCD_BLUE
LCD_GREEN
LCD_RED
DDC_SCL
LCD_VSYNC_5V
LCD_HSYNC_5V
DDC_SDA
C101
33pF/50V
L4
1uH/80mA
C100
100pF/25V
C103
270pF/50V
LCD_BLUE_N
LCD_BLUE
Micro SD Card
De-Cap
0.1uF/16V/X7R
C112
10uF/10V/X5R
C113
FB9
MMZ2012S121A
GNDD
J8
Micro_SD_Card_Det1
MMCSD0_D2
MMCSD0_D3
MMCSD0_CMD
MMCSD0_CLK
MMCSD0_D0
MMCSD0_D1
GPIO4[0]
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET
SHIELD
SH
Pull-Up
10K,1%,1/16W
R91
R92
R93
R94
R95
R96
R97
Logsun Systems
Sheet: /SD_Card_LCD/
File: SD_Card_LCD.kicad_sch
Title: DSP320C6748 DVK
Size: A4
Date: 2026-03-28
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 8/17

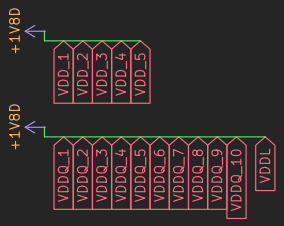

IC11
THS8135PHP
IC13
SN74AVC16T245DGG
IC12
SN74AVC1T45DBVR
U6
TXS0102DCU
U7
SN74LVC2T45DCUR
+3V3ADAC
+1V8D
+3V3D
GND
VDAC_AGND
0.1uF/16V/X7R
LCD_AC_ENB_CS_1V8
LCD_PCLK_1V8
LCD_HSYNC_1V8
LCD_VSYNC_1V8
LCD_HSYNC
LCD_VSYNC
LCD_GREEN_N
LCD_RED_N
LCD_BLUE_N
I2C0_SCL
I2C0_SDA
TS_INT
GPIO2[6]
Logsun Systems
Sheet: /LCD_II/
File: LCD_II.kicad_sch
Title: DSP320C6748 DVK
Size: A4
Date: 2026-03-28
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 9/17









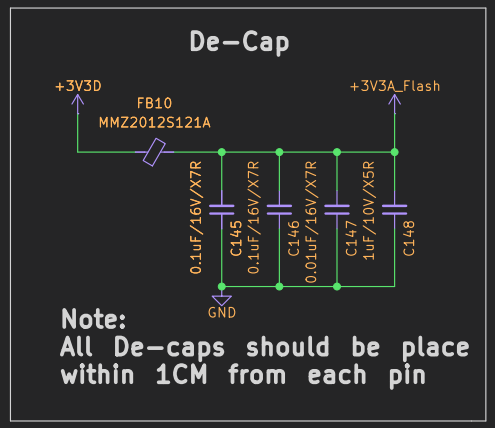


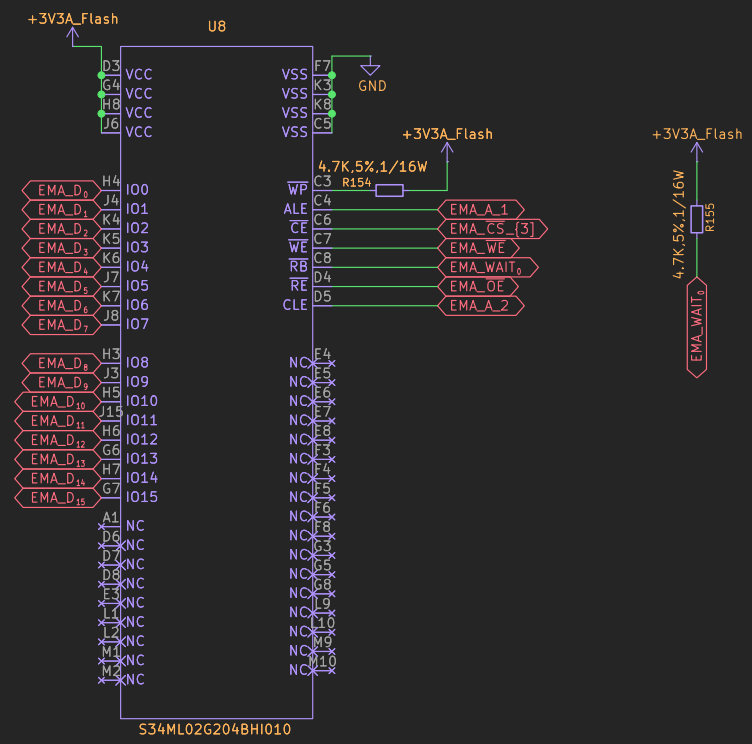
Logsun Systems

Sheet: /DDR_RAM/
File: DDR_RAM.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 10/17



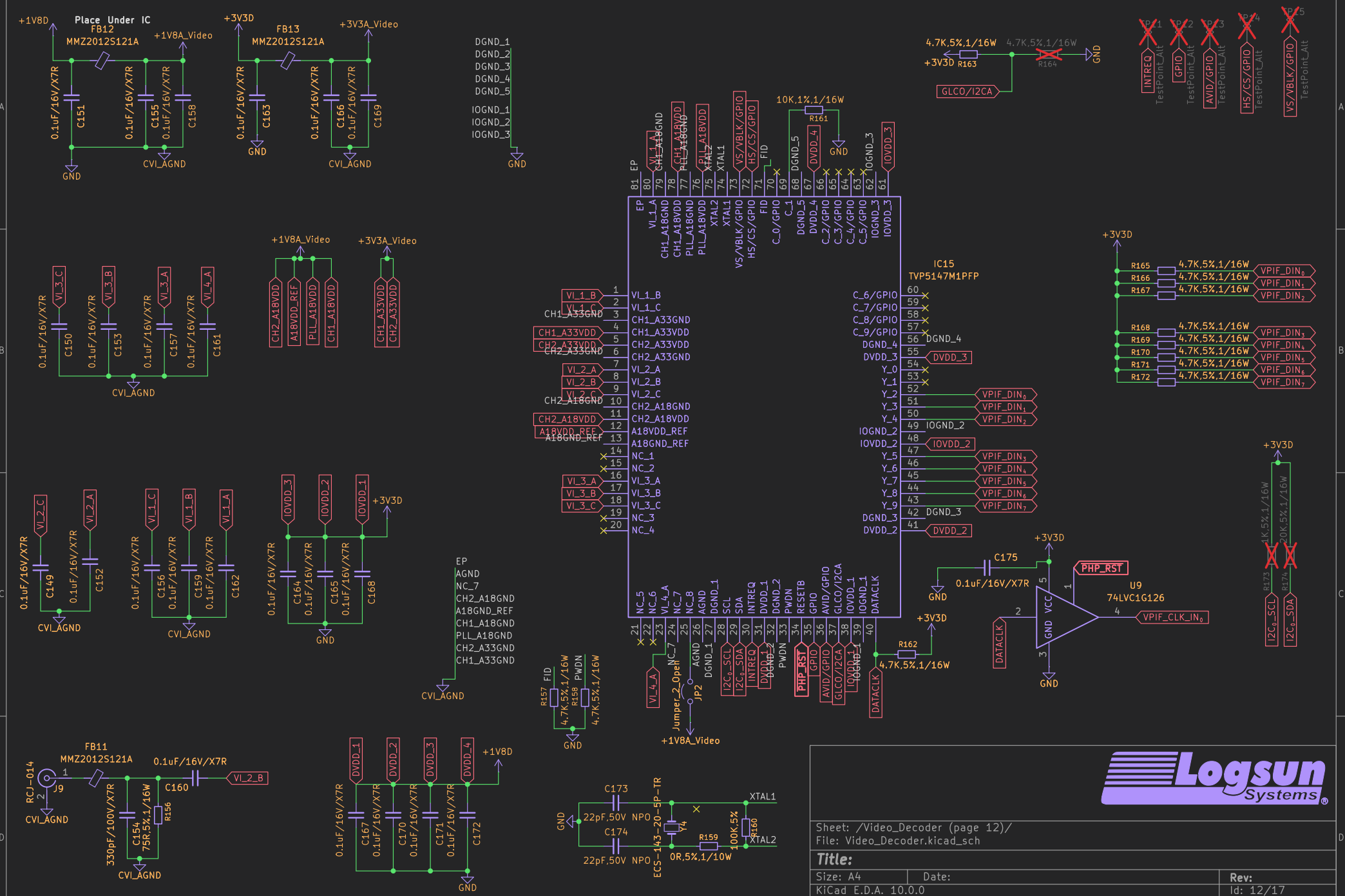


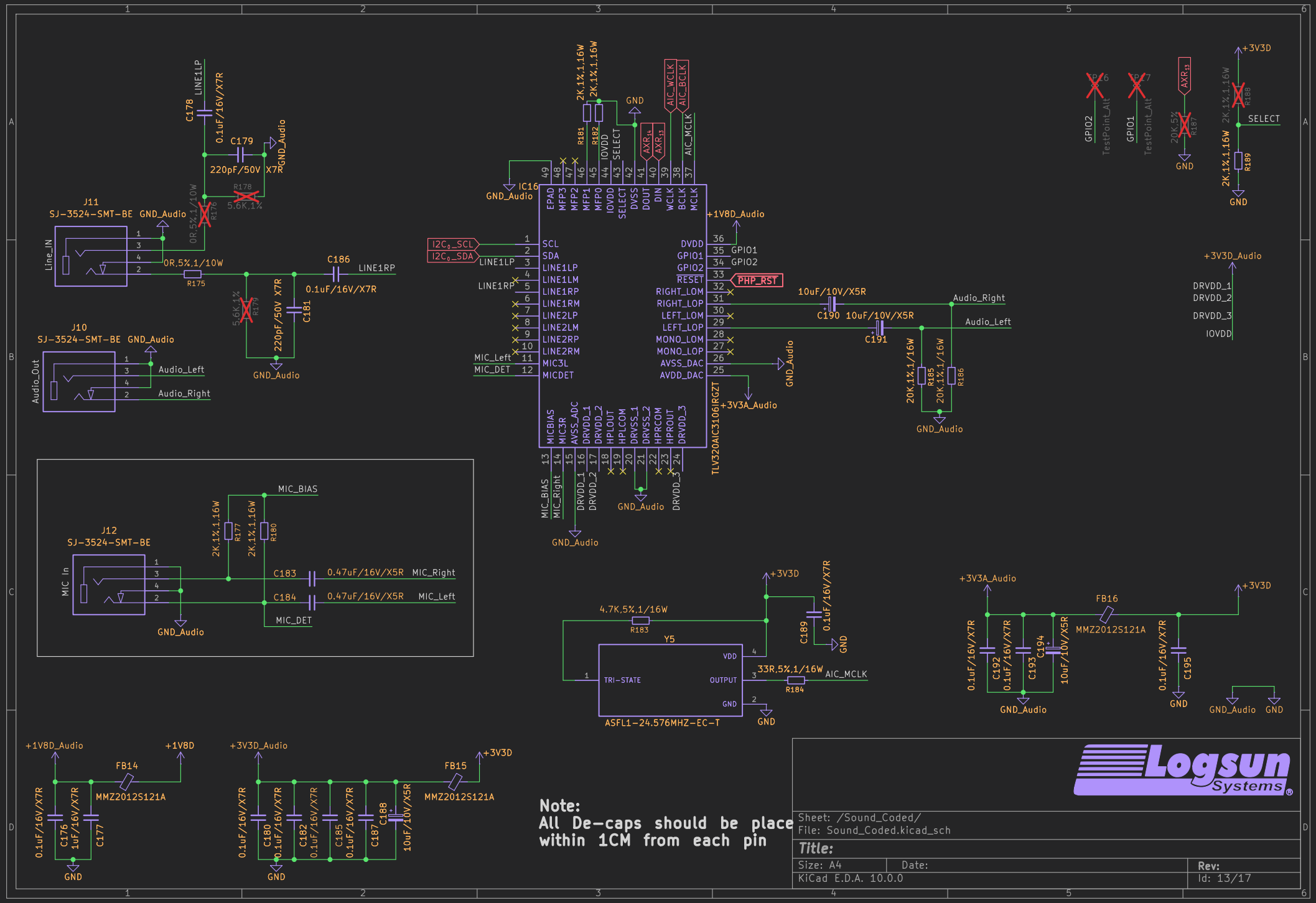
Logsun Systems

Sheet: /NAND_Flash/
File: NAND_Flash.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 11/17


Place Under IC
FB12
MMZ2012S121A
FB13
MMZ2012S121A
+1V8D
+1V8A_Video
+3V3D
+3V3A_Video
0.1uF/16V/X7R
C151
C155
C158
C163
C166
C169
GND
CVI_AGND
DGND_1
DGND_2
DGND_3
DGND_4
DGND_5
IOGND_1
IOGND_2
IOGND_3
4.7K,5%,1/16W
R163
R164
GLCO/I2CA
INTREQ
GPIO
AVID/GPIO
HS/CS/GPIO
VS/VBLK/GPIO
TestPoint_Alt
10K,1%,1/16W
R161
IC15
TVP5147M1PFP
C150
C153
C157
C161
CH2_A18VDD
A18VDD_REF
PLL_A18VDD
CH1_A18VDD
CH1_A33VDD
CH2_A33VDD
C149
C152
C156
C159
C162
C164
C165
C168
IOVDD_3
IOVDD_2
IOVDD_1
EP
AGND
NC_7
CH2_A18GND
A18GND_REF
CH1_A18GND
PLL_A18GND
CH2_A33GND
CH1_A33GND
R157
R158
FID
PWDN
JP2
Jumper_2_Open
R165
R166
R167
R168
R169
R170
R171
R172
VPIF_DIN0
VPIF_DIN1
VPIF_DIN2
VPIF_DIN3
VPIF_DIN4
VPIF_DIN5
VPIF_DIN6
VPIF_DIN7
DVDD_3
DVDD_2
IOVDD_2
C175
PHP_RST
U9
74LVC1G126
VPIF_CLK_IN0
DATACLK
R162
R173
R174
I2C0_SCL
I2C0_SDA
1K,5%,1/16W
20K,5%,1/16W
RCJ-014
J9
FB11
MMZ2012S121A
C160
VI_2_B
330pF/100V/X7R
C154
75R,5%,1/16W
R156
C167
C170
C171
C172
DVDD_1
DVDD_2
DVDD_3
DVDD_4
C173
C174
22pF,50V NPO
ECS-143-20-5P-TR
Y4
R159
0R,5%,1/10W
100K,5%
R160
XTAL1
XTAL2
Logsun Systems
Sheet: /Video_Decoder (page 12)/
File: Video_Decoder.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.0
Id: 12/17


J11
SJ-3524-SMT-BE
GND_Audio
Line_IN
C178
LINE1LP
0.1uF/16V/X7R
C179
220pF/50V X7R
GND_Audio
R178
5.6K,1%
R176
0R,5%,1/10W
0R,5%,1/10W
R175
C186
LINE1RP
0.1uF/16V/X7R
5.6K,1%
R179
220pF/50V X7R
C181
GND_Audio
J10
SJ-3524-SMT-BE
GND_Audio
Audio_Out
Audio_Left
Audio_Right
IC16
GND_Audio
2K,1%,1.16W
R181
R182
IOVDD
SELECT
GND
AIC_WCLK
AIC_BCLK
AXR14
AXR13
AIC_MCLK
EPAD
MFP3
MFP2
MFP1
MFP0
IOVDD
SELECT
DVSS
DOUT
DIN
WCLK
BCLK
MCLK
I2C0_SCL
I2C0_SDA
LINE1LP
LINE1RP
MIC_Left
MIC_DET
SCL
SDA
LINE1LP
LINE1LM
LINE1RP
LINE1RM
LINE2LP
LINE2LM
LINE2RP
LINE2RM
MIC3L
MICDET
DVDD
GPIO1
GPIO2
RESET
RIGHT_LOM
RIGHT_LOP
LEFT_LOM
LEFT_LOP
MONO_LOM
MONO_LOP
AVSS_DAC
AVDD_DAC
+1V8D_Audio
GPIO1
GPIO2
PHP_RST
10uF/10V/X5R
C190
10uF/10V/X5R
C191
Audio_Right
Audio_Left
GND_Audio
+3V3A_Audio
TLV320AIC3106IRGZT
20K,1%,1/16W
R185
20K,1%,1/16W
R186
GND_Audio
MICBIAS
MIC3R
AVSS_ADC
DRVDD_1
DRVDD_2
HPLOUT
HPLCOM
DRVSS_1
DRVSS_2
HPRCOM
HPROUT
DRVDD_3
MIC_BIAS
MIC_Right
DRVDD_1
DRVDD_2
GND_Audio
GND_Audio
DRVDD_3
GPIO2
TestPoint_Alt
GPIO1
TestPoint_Alt
AXR13
20K,5%
R187
GND
+3V3D
2K,1%,1.16W
R188
SELECT
2K,1%,1.16W
R189
GND
+3V3D_Audio
DRVDD_1
DRVDD_2
DRVDD_3
IOVDD
J12
SJ-3524-SMT-BE
MIC In
GND_Audio
2K,1%,1.16W
R177
2K,1%,1.16W
R180
MIC_BIAS
C183
0.47uF/16V/X5R
MIC_Right
C184
0.47uF/16V/X5R
MIC_Left
MIC_DET
+3V3D
4.7K,5%,1/16W
R183
C189
0.1uF/16V/X7R
GND
Y5
VDD
TRI-STATE
OUTPUT
GND
33R,5%,1/16W
R184
AIC_MCLK
GND
ASFL1-24.576MHZ-EC-T
+3V3A_Audio
FB16
MMZ2012S121A
+3V3D
0.1uF/16V/X7R
C192
0.1uF/16V/X7R
C193
C194
10uF/10V/X5R
GND_Audio
0.1uF/16V/X7R
C195
GND
GND_Audio
GND
+1V8D_Audio
FB14
MMZ2012S121A
+1V8D
0.1uF/16V/X7R
C176
1uF/16V/X7R
C177
GND
+3V3D_Audio
FB15
MMZ2012S121A
+3V3D
0.1uF/16V/X7R
C180
0.1uF/16V/X7R
C182
0.1uF/16V/X7R
C185
0.1uF/16V/X7R
C187
C188
10uF/10V/X5R
GND
Note:
All De-caps should be place
within 1CM from each pin
Logsun Systems
Sheet: /Sound_Coded/
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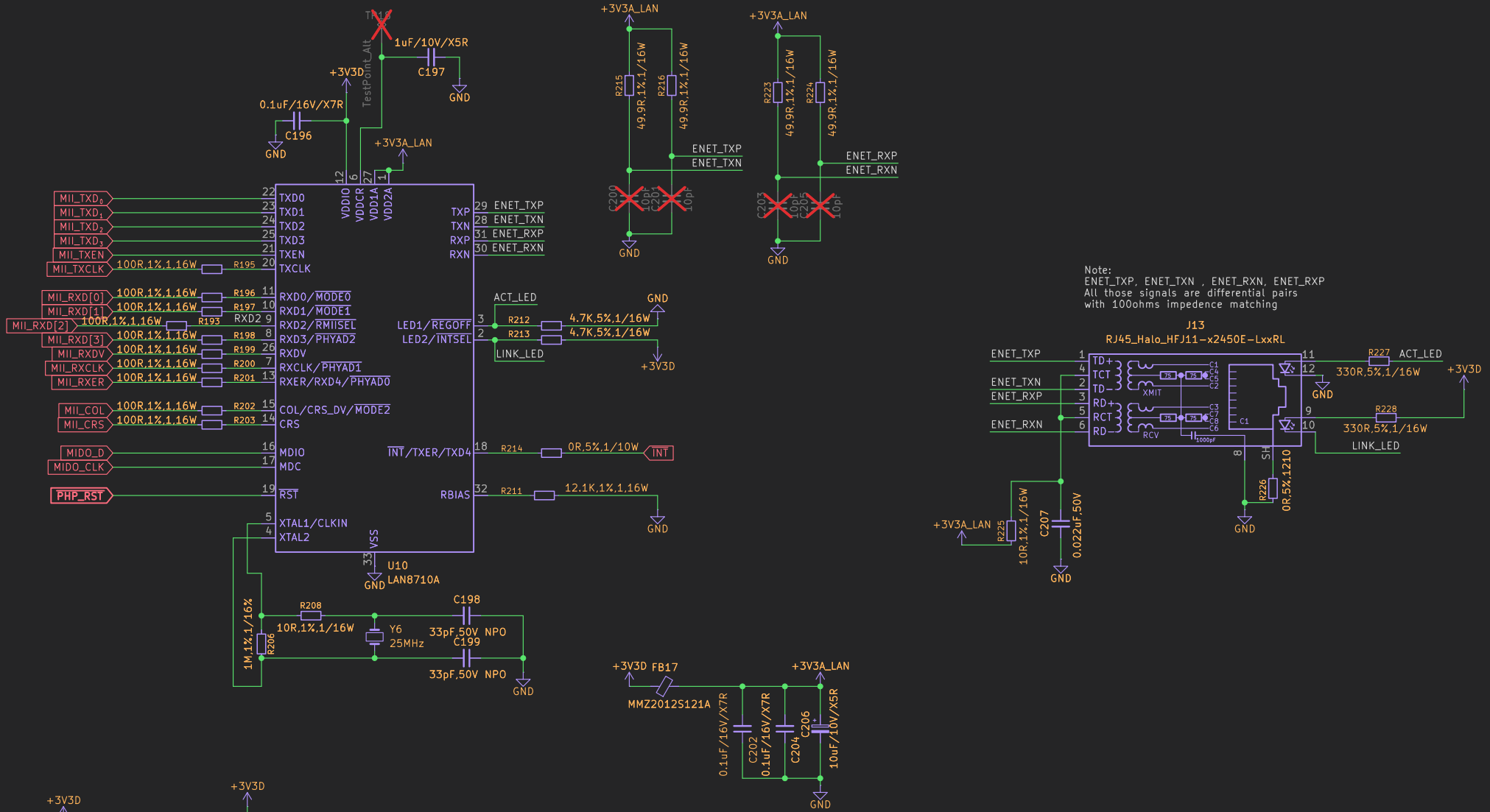
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.0
Id: 13/17



Logsun Systems

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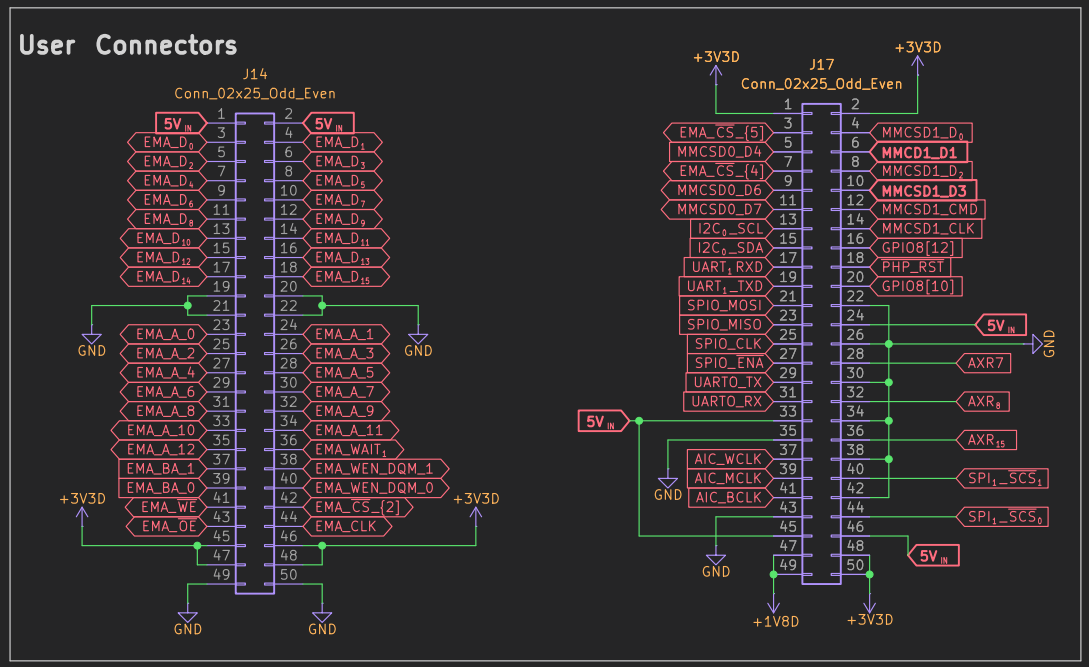
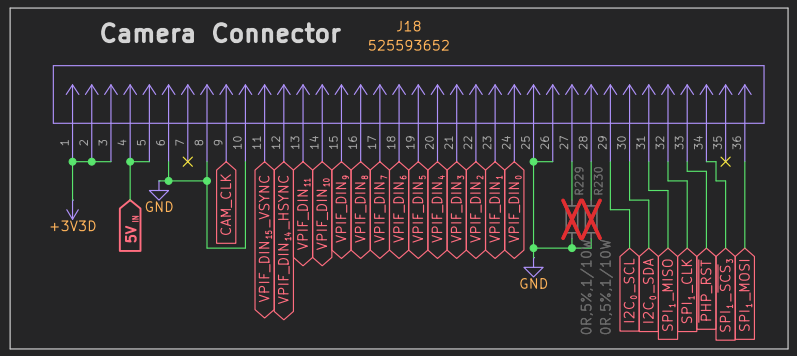
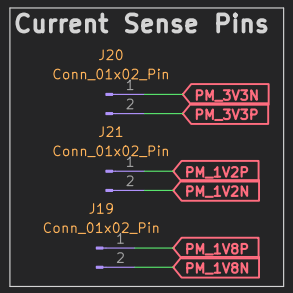
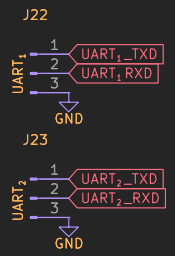
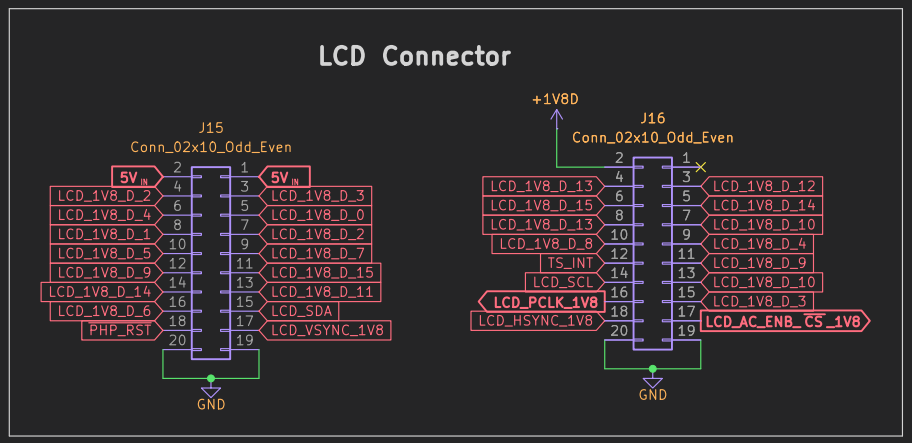
KiCad E.D.A. 10.0.0		Id: 14/17











GPIO0[8]

Logsun Systems

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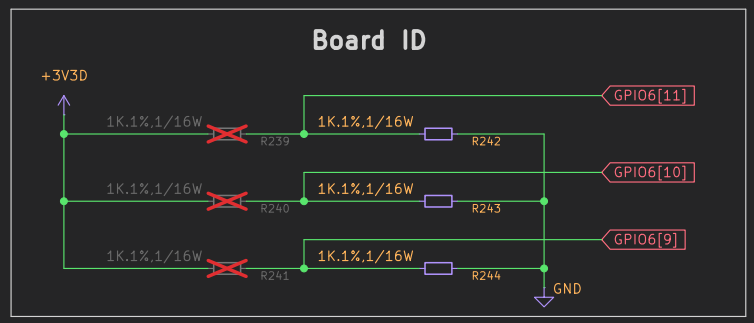
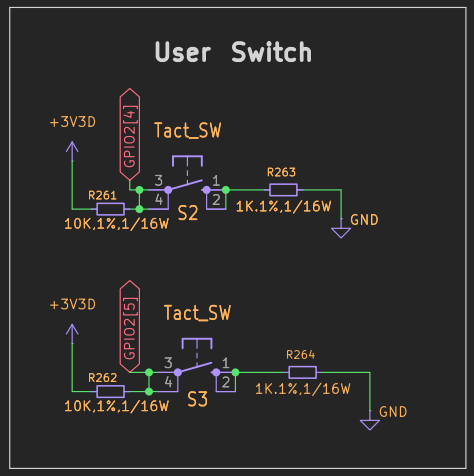
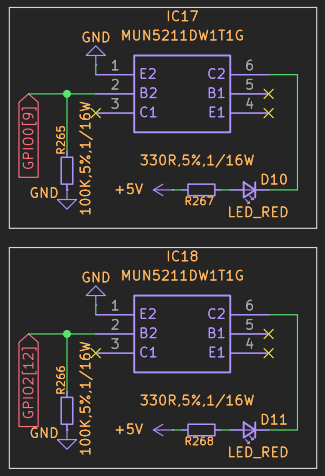
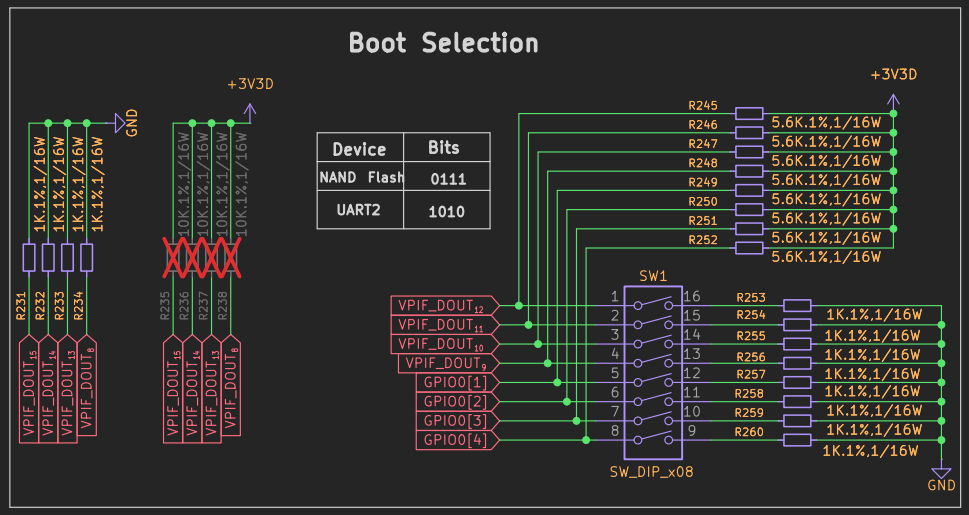
KiCad E.D.A. 10.0.0		Id: 15/17









Logsun Systems

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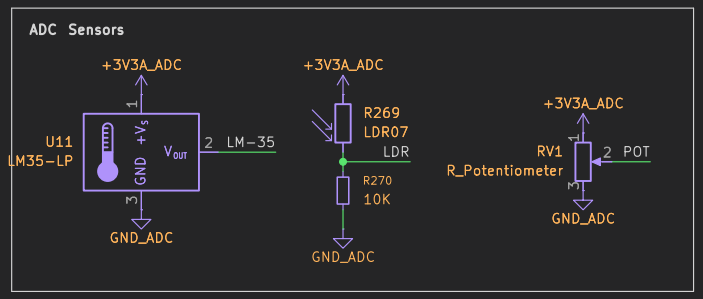
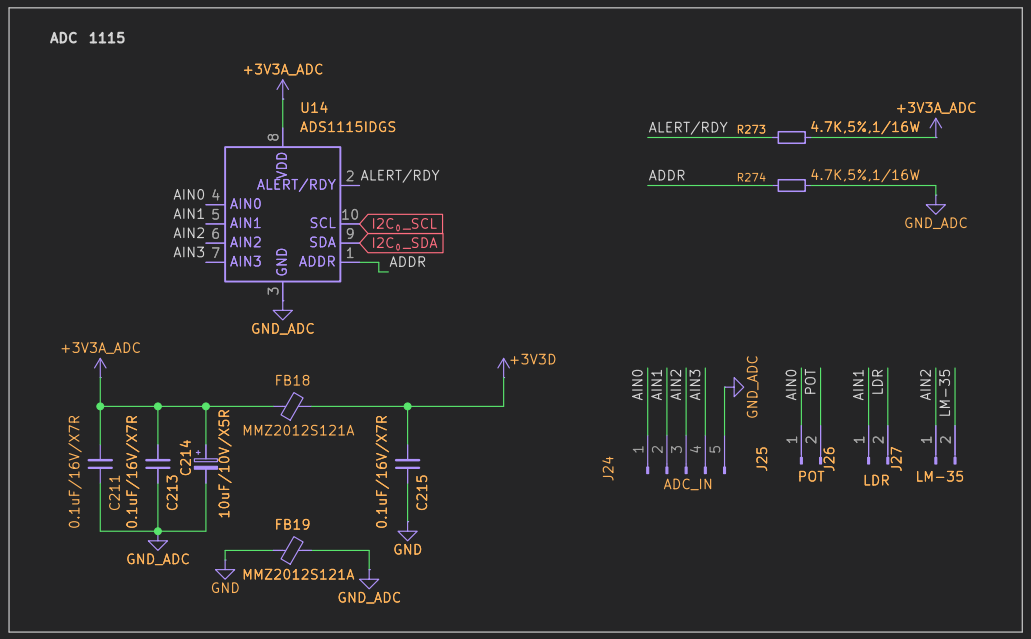
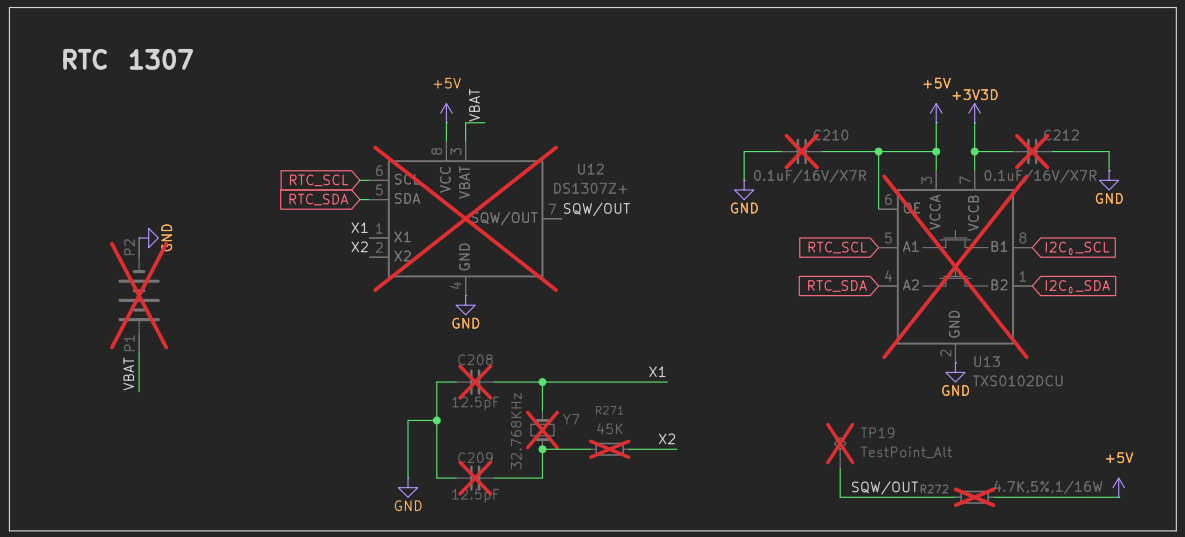
KiCad E.D.A. 10.0.0		Id: 16/17







Logsun Systems

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